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Multi-Purpose Deployable Systems and Capability Development Platform: Preparation and Early Operations of the TPA-1 CubeSat

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Abstract

TPA-1, a 3U CubeSat developed by the University of Auckland's Te Pūnaha Ātea - Space Institute (TPA-SI), was successfully launched on 23rd June 2025. The mission serves as a capability development platform, demonstrating solutions to common nanosatellite challenges such as in-orbit inspection and sustainable end-of-life disposal, while simultaneously validating the TPA-SI's end-to-end mission delivery processes from our New Zealand-based facilities and team. The primary technology is a 1 m restowable, deployable composite boom hosting a stereoscopic inspection camera. In-orbit operations have verified its unique multi-cycle deployment and retraction capability, a significant advance over single-shot systems. Its tip-mounted cameras provide detailed imagery for spacecraft health monitoring, enabling assessment of material degradation and the identification of a suspected micrometeoroid (MMOD) impact on a solar cell. A second key technology is a compact, "Tuna-can-sized" dragsail system that fits entirely within the extended volume of standard CubeSat deployers, preserving valuable internal spacecraft volume for primary payloads. This low-resource deorbit solution has proven its commercial viability, forming the basis of a spin-out company now developing larger systems for the microsatellite market. The primary operational payloads include an Earth observation imager and a dedicated deployables imager, both built on accessible Raspberry Pi platforms. These systems have returned high-quality imagery of both Earth and the spacecraft's own deployed structures, serving as a flexible, Python-based testbed for students and researchers to develop and validate on-board image processing algorithms. This paper summarises the successful on-orbit performance of these payloads, presenting initial operational results that confirm the achievement of primary mission objectives and showcase the platform's layered utility for advanced technology demonstration, hands-on education, and future commercial applications.

Keywords: In-orbit demonstration, Earth Observation, CubeSat, End-of-Life disposal, Capability Development

Acronyms/Abbreviations

BRC - Bistable Rollable Composite
CTS - Constant Torque Spring
LEOP - Launch and Early Operations Phase
MMOD - MicroMeteorite and Orbital Debris
MOCC - Mission Operation and Control Centre
NFI - Narrow Field Imager
NSTF - National Satellite Test Facility
RTL - RolaTube Technology Ltd.
TMU - Thermal Monitoring Unit
TPA-SI - Te Pūnaha Ātea - Space Institute
USRP - Universal Software Radio Peripheral
WFI - Wide Field Imager

1 Introduction

1.1 Te Pūnaha Ātea– Space Institute

To support the growth of the New Zealand space sector and coordinate space activities within the organization, the University of Auckland established Te Pūnaha Ātea – Space Institute (TPA-SI) in 2019. TPA-SI is a multidisciplinary centre of expertise in space science and engineering across the faculties of Engineering and Science. TPA-SI has established capability to execute space missions and develop applications. The group carries out applied research in space science and engineering and runs educational activities that shape the next generation of scientists and engineers. To support these activities, TPA-SI has established national level facilities, including an ISO Class 7 cleanroom, the National Satellite Test Facility (NSTF) with full launch and space environment qualification facilities, and a Mission Operations Control Centre (MOCC), an enduring piece of national infrastructure and subsequently other homegrown and international missions.

TPA-SI is now established as a key player in the national space ecosystem, also with several international collaborations ongoing across Europe, the US and Asia-Pacific.

1.2 Capability development Programme

An initial series of two CubeSat missions are underway to progressively build capability and capacity

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within TPA-SI to allow future delivery of reliable, application-led missions. Whilst many TPA-SI personnel have delivered missions in the past, the TPA-1 CubeSat mission is being used to validate the processes, facilities and planning for conducting missions from New Zealand with its own distinct supply chain and regulatory environment.

The first mission from TPA-SI is intended to establish the process for flight hardware development and end-to-end mission delivery, using a commercial-off-the-shelf avionics platform from an established supplier. The spacecraft is a 3U CubeSat hosting payloads with outreach potential and as test cases for working with domestic and international partners. This mission, along with the follow up, TPA-2, is intended to help "complete the picture" of a national capability for small scale missions for academic institutions, startups and spinouts, as well as forming a first step towards larger scale national missions. A key element of this programme is its potential for education and outreach potential. Consisting of domestically integrated satellites owned by a New Zealand entity, the intention is to use these as eductional teaching aids at Universities and schools across the country to connect young people with space.

The payload development [1] and integration activities [2] towards this mission have previously been reported at previous conferences.

1.3 Paper Structure

The paper summarises the four payloads developed for the mission, their performance in orbit, plans for ongoing operations and their wider application. Section 2 outlines the overall mission as well as each of the payloads. Section 3 discusses final preparation through Launch and Early Operations Phase (LEOP), Section 4 discusses the payload performance and Section 5 discusses future planned operations and further applications.

2 TPA-1 Mission Overview

2.1 Motivation

The payload suite development for the mission was driven by two key considerations 1) outreach and education potential, and 2) foundational capabilities to enable future application and science driven missions.

As such, six cameras have been integrated into the mission, including pairs in a stereoscopic configuration. The mission is intended to develop skills in Earth Observation operations and to demonstrate versatile

operations with deployable structures (in this case, a Boom) with application in instrumentation, solar arrays and antenna systems. Finally, as good practice for a sustainable space programme, a dragsail is utilised to dispose of the spacecraft in a timely fashion.

2.2 Mission Success Criteria

At inception, the following mission success criteria were established:

Minimum mission success criteria

- Spacecraft deployment and signal acquisition in orbit
- Bi-directional link to spacecraft established via TPA-SI ground facilities
- Activation and data downlink of at least one imaging payload

Full Mission Success criteria

- Identifiable imagery captured of NZ landmass
- Downlink imagery of deployed Boom system
- Performance data for thermal payload captured
- Deployment of dragsail system
- Downlink imagery of deployed dragsail

In the first two months of operation, all success criteria barring those related to the deployment of the dragsail have been met.

2.3 Platform overview

As a first mission, TPA-SI elected to procure a platform from an established provider. Nanoavionics was selected to supply the two 3U CubeSat buses for TPA-1 and TPA-2. The 3U buses have high performance as reflected in Table 1. A bespoke configuration was designed between Nanoavionics and TPA-SI to host the planned payloads with appropriate cutouts in fixed and deployable panels. Two deployable solar arrays provide 14 GaAs cells on the +X facet, in addition to body panel mounted cells around the structure. Ground communications can be conducted via omnidirectional canted turnstile UHF, or directional S-band patch antennas.

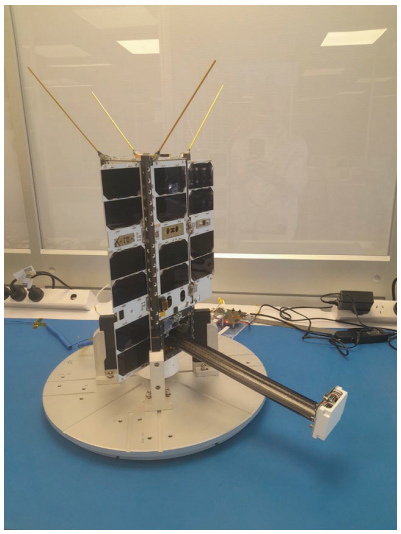


Figure 1: TPA-1 with Boom partially deployed

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Table 1: TPA-1 Platform capabilities

Parameter	Performance
Power generation and storage	-13 W sun-pointing generation -40 Whr Li-ion battery (4 cell)
Uplink/Downlink	-UHF (402MHz) 2.4-19.2 kbps GFSK2; -S-band (2.2GHz) 128/500kbps (up/down) GMSK
Position and Attitude	-3-axis pointing control (2°/5° sunlight/eclipse); -10 m position knowledge
Payload Data Storage	-256 Mbytes (Flash) + 2 GBytes (SD card) payload data, split across 10 data slots

2.4 Imagers Payload

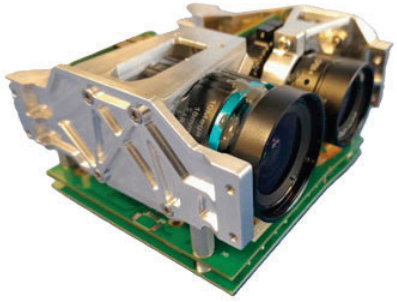


Figure 2: TPA-1 Imagers Payload

The imagers payload consists of two independent payloads integrated on a common interface PCB and mechanical structure.

The Earth Observation imager payload uses an off-the-shelf Raspberry Pi compute module 4 mounted on a bespoke interface board and connected to two camera systems. The EO imager interfaces to the spacecraft via power and UART serial connections. The camera systems are off-the-shelf Raspberry Pi cameras with optics ruggedized for flight [1]. The Narrow Field Imager (NFI) and Wide Field Imager (WFI) cameras provide 50 m and 250 m ground resolution at a maximum frame of 12.3 MP (4056x3040) using a Sony IMX477R sensor through 16 mm and 6 mm focal length lenses, respectively with a field of view of 27.4 degrees and 76.3 degrees across the diagonal.

The Raspberry Pi module is in a headless configuration allowing for commanding over UART. The payload has full interface functionality allowing for upload of new control and image processing python scripts in orbit. The payload is designed as an accessible hardware and software environment for the development of on-board image processing tools for students and researchers to trial applications in-orbit.

The EO imager is pointed through a side panel cutout on the anti-sun/nadir facet (-X) of the spacecraft.

The Deployables imager payload is of the same basic hardware and software configuration as the EO imager payload, but here configured as a stereoscopic imager with 8 MP wide field of view cameras and a focal distance of ~20 cm to image both the boom and dragsail deployments. The imager is angled at 70 degrees from normal to contain as much of the deployment in Field of view of the cameras. The deployables imager is pointed through a side panel cutout on the sun pointing/ zenith facet (+X) of the spacecraft.

2.5 Thermal Monitoring Unit

The thermal monitoring payload provides readout from multiple thermal sensors located around the spacecraft managed by an MSP430FR microcontroller, supplementing that available from integrated subsystem sensors. The data generated will be used to develop and verify thermal models for future missions.

Six LMT85 external temperature sensors are distributed on the spacecraft, with two positioned to support a materials experiment from Cirrus Materials Science, and others placed to support payload operations. The payload will be active throughout nominal operations, with data logged to the Payload Computer every 60 seconds for later downlinking.

2.6 Inspection Boom

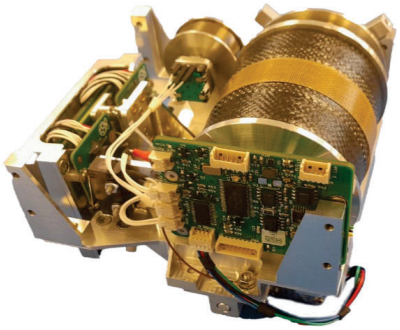


Figure 3: TPA-1 Inspection Boom Payload

The Inspection boom system consists of a deployable Bi-stable Rollable Composite (BRC) boom and motor driven deployment mechanism, associate electronics and a Raspberry Pi based imaging system. The boom has a deployable length of 1 m and is driven by a 12V stepper motor controlled from a dedicated motor controller board with encoder position feedback. The boom can be deployed and stowed by command over CAN to an MSP430FR microcontroller and an L6470 motor controller. An integrated dyneema burnwire system retains the boom tip module during launch. The boom is manufactured by Rolatube Technology Ltd. from carbon fibre-epoxy plies that are arranged to produce bistability[3].

The deployment is driven indirectly by a stepper motor via a drive ribbon that is co-coiled with the boom. With the drive ribbon attached to the outside face of the boom, the ribbon not only rotates the deployment drum, but also applies a radial pressure to the boom to prevent blossoming. Whilst a number of inspection booms are

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available on the market [4,5], a distinguishing feature of this system is that the boom can be repeatedly deployed and stowed, as well as automatically retracted in the event of power failure to the unit. To enable stowing a Constant Torque Spring (CTS) is attached to the deployment drum so that the motor simultaneously deploys the boom and winds the CTS onto the drum. The motor holds the boom in the deployed state against the torque of the CTS. Upon stowing, the motor acts as a break as the CTS stows the boom. Should there be an anomaly cutting power to the system when the boom is extended the CTS automatically retracts the boom.

2.7 Dragsail

The dragsail payload embarked on TPA-1 is designed to make use of the extended volume of the CubeSat form factor, informally described as the "Tuna-Can". The mechanism is designed to be compatible with the majority of CubeSat deployment pods, including those produced by ISISpace, Exolaunch, with a diameter of 79mm and height of 43mm. The hold-down release mechanism consists of a deployment dyneema burnwire releasing a compression spring, lifting a boom and sail compartment. The burnwire is tensioned by an extension spring with mechanical advantage by multiple passes of the dyneema string between the deployable section and retaining base. Once released, spring steel booms are then deployed laterally, which pull thin metalised film sail quadrants to a 1 m^2 total area. The dragsail mechanism will be deployed by command via a switched 3V3 power line. The dragsail contains a recess and feed through path for integration of the ADCS -Z Sun sensor as seen in Figure 4.

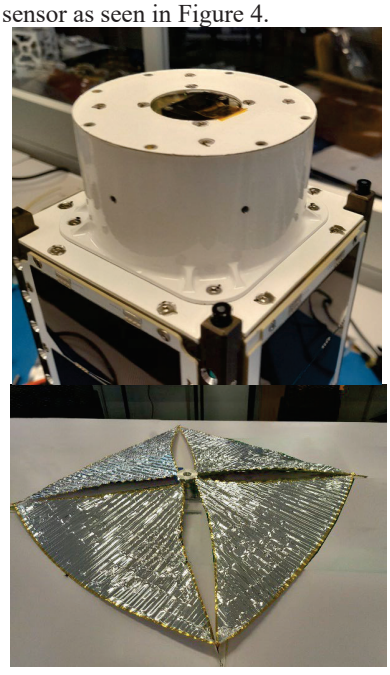


Figure 4: TPA-1 NANO dragsail in stowed (top) and deployed (bottom) states

2.8 Ground Segment

Mission operations are conducted from the TPA-SI Mission Operations and Control Centre (MOCC) located in the Faculty of Engineering at the University of Auckland. The primary ground station is situated at the university's Ardmore field site (37.03S, 174.99E). The station features co-located UHF antennas and a site to establish a steerable S-band antenna, controlled by Universal Software Radio Peripheral (USRP) devices running GNUradio flowgraphs. The mission control software is initially based on the NanoMCS from Nanoavionics, with operations transitioning to OpenC3 COSMOS platform to provide command, control, and monitoring of the spacecraft [6]. This architecture supports both local and remote operations and is designed to be scalable and rapidly configurable for future missions.



Figure 5: TPA-SI ground segment facilities: Mission Operations and Control Centre (MOCC) (top), Ardmore ground station (bottom).

3 Preparation and Launch

3.1 Final Integration

Subsystems testing is reported in [Small Sat]. All payloads were subjected to environmental testing including active thermal vacuum and vibration testing prior to final integration into the spacecraft platform. Limited over the air testing was conducted with an attenuated cleanroom USRP replicating the

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modulation/demodulation of the radio signals. Primary testing was conducted via spacecraft umbilical.

3.2 Environmental qualification

The spacecraft was tested to Protoflight levels in accordance with Falcon 9 requirements [7] following a sequential test flow of vibration testing followed by thermal vacuum testing conducted in full at the NSTF located at TPA-SI.

The spacecraft was integrated into a testpod with deployables prepped as per flight after a full functional bench test. Three axis testing was then performed to Falcon 9 random vibration loads at $7.87 g_{rms}$ random vibration with pre/post test sine sweeps. A post vibe functional test then confirmed full functionality of the system, and that there was no inadvertent activation of the power system by verification of boot counts.

The spacecraft was then mounted in the Thermal Vacuum chamber with umbilical and RBF interfaces routed to external ports. Thermocouples mounted to the +/-Z and -Y facets of the spacecraft provided temperature reference throughout cycling. Clearance was confirmed with the antenna, solar panels and boom deployables. The spacecraft was cycled between -10 °C and +60 °C demonstrating both cold and hot turn on with antenna and solar deployments triggered at cold.

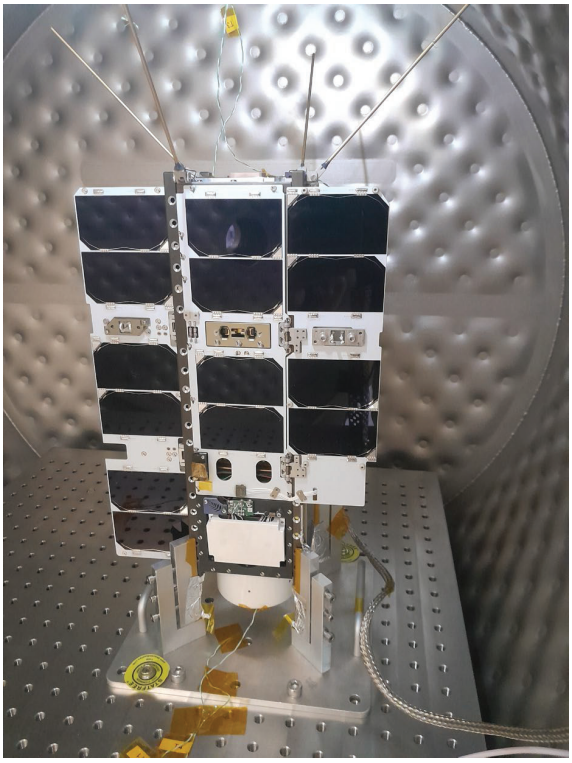


Figure 6: TPA-1 at the end of thermal vacuum testing at the TPA-SI NSTF

3.3 Flight rehearsal and Day In The Life

A 48-hour rehearsal test was conducted from deployment from a spacecraft test-pod through the first two days of operation with flight-like on board configurations. The spacecraft was then operated over the air as it would be for flight including automated power up antenna deployment, beacon transmissions and entering into a detumbling mode. The spacecraft was then commanded through initial commissioning sequence through to activation of the payloads through pre-loaded on-board scripts. The spacecraft was powered intermittently via umbilical and onboard reset watchdogs allowed to expire with recovery back to nominal state practiced.

3.4 Launch and Early Operations Phase

The satellite was shipped for integration to ISISpace in Delft, Netherlands in April 2025. Final preparation included inspection of the spacecraft on arrival, removal of all Remove Before Flight items. Final flight configurations including timing for deployments of solar arrays and antennas were confirmed, pre-loaded payload scripts verified and counters were recorded.

Fit-checks and spacecraft insertion and deployment from the Quadpack dispenser were conducted three times to verify smooth deployment and shutdown/ activation of the spacecraft kill-switches.

After integration, the Quadpack hosting the TPA-1 satellite was delivered to launch site for integration with the launch vehicle. Transporter-14 launched from Vandenberg SFB, California on 23rd June, with TPA-1 deploying at T+55 minutes.

Initial contact was made with the spacecraft on the first pass over the Ardmore ground-station with beacon detection. Over the following days and contacts, initial telemetry decoding was performed and power and thermal health of the spacecraft verified, as well confirmation of low and reducing tumbling rates.

Refinement of communications improved half duplex TT&C and allowed downlink of spacecraft files, setting the spacecraft to 9k6 baud transmissions to aid throughput.

The spacecraft was transitioned to pointing mode, nominally sun pointing with GEO pointing at the Ardmore ground-station taking precedence during passes.

4 Mission Operations

4.1 Attitude control

For the first two months of the mission, the spacecraft maintained sun-pointing attitude to maximise power generation. It was noted that whilst the spacecraft power budget held significant margin, the drag on the spacecraft resulted in an drop in altitude averaging ~46 m/day from a 514 km starting altitude. Conscious of increasing solar activity and consequent increase in rate of decent, the spacecraft was transitioned to a velocity

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vector pointing mode such that the minimum area +Z facet is presented to the ram direction, with the -X facet hosting the EO imager cameras pointing towards Nadir. Assessment of the spacecraft power budget demonstrated a healthy margin. Figure 7 shows the B* drag term from TLEs from TPA-1 and ConnectaIoT-9, a 6U CubeSat co-manifested on the Transporter-14 launch. Here, the transition of TPA-1 to a low drag profile is clearly indicated occurring on day 240.

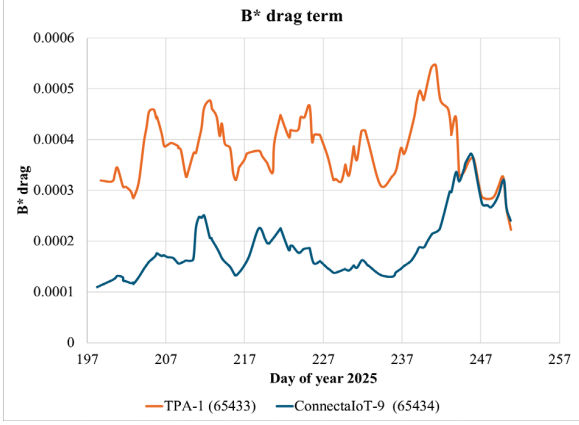


Figure 7: History of B* drag term from TLEs of TPA-1 and ConnertaIoT-9

4.2 Thermal Monitoring Unit (TMU)

The TMU was the first payload activated during the mission and has been active near continuously gathering thermal data used to support operations of other payloads and verify spacecraft safety.

Figure 8 shows representative data from two orbits largely in sun-pointing mode. The dip observed in external sun facing sensors at ~22:50 is due to a reorientation to point at the Ardmore ground station during a pass.

4.3 Imaging operations

After verification of spacecraft health, the imagers payload was activated and initial low-quality jpeg imagery captured to verify functionality and downlink capability. Update of scripts improved adjusted quality settings to a compromise between jpeg compression artifacts and file size for download.

Imaging operations were standardised to downlink the first ~10 kB of image files containing a thumbnail to assess image quality and determine if a full image download is warranted.

Initial imagery was captured during pointing at the Ardmore groundstation. Alternate scripting was uploaded to generate reduced resolution imagery such that multiple images could be captured and stacked in the available slots on the spacecraft payload computer. In general, images are captured from both the WFI and NFI imagers, with the NFI occupying the centre frame.

Multiple image captures have been performed to date, targeting other regions of interest both during overhead passes and off-nadir pointing. Several imaging targets have been selected by student interns on a technical basis for regions of high contrast and image complexity.

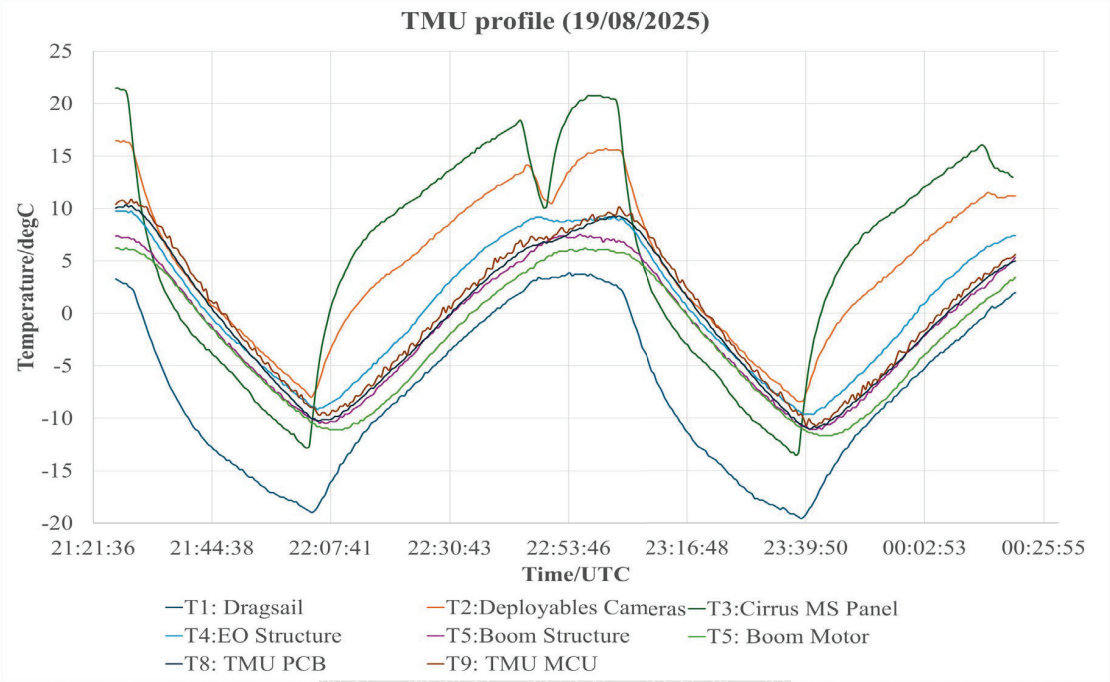


Figure 8: Representative thermal data as captured by the TMU

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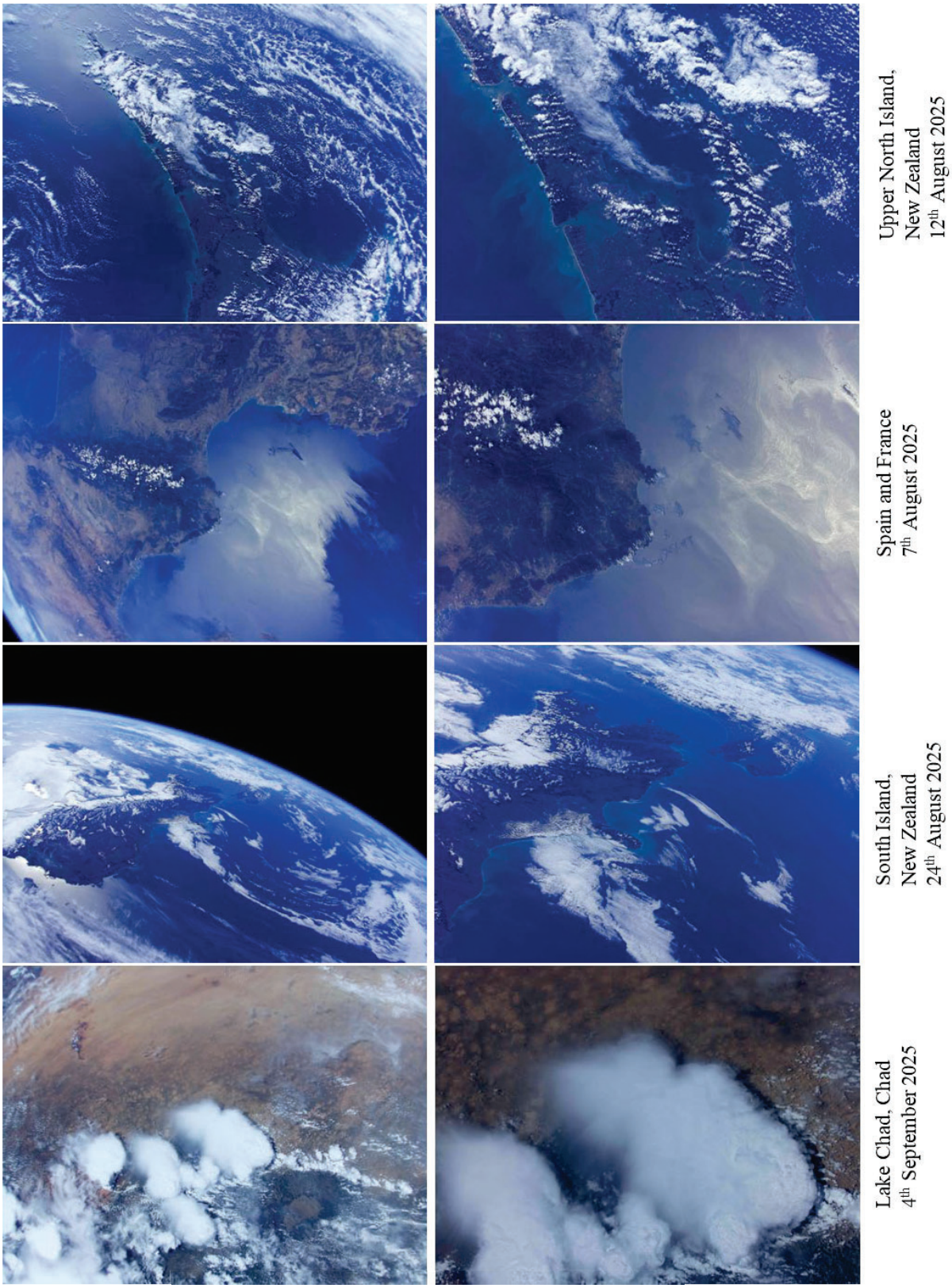


Figure 9: Subset of Imagery captured by TPA-1 EO WFI (left) and NFI (right)

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4.4 Boom deployments

The Inspection boom payload is retained for launch via a Dyneema burnwire system to restrain movement of the boom tip under launch loading. An initial script sequence performed activated the burnwire circuit built into the boom. A second script was later executed to perfom a 5 mm movement of the boom with images captured from the boom tip camera before and after the minimal deployment, before performing a retraction. This test verified that the boom was now free to be deployed after severing of the burnwire.

Pre-set scripting was set to perform a 25 cm deployment and capture an image from each boom tip camera at low quality. This script was executed over Antarctica on 21st August 2025.

A second deployment test was conducted to 38 cm after the spacecraft transitioned to velocity vector/nadir pointing during a pass over New Zealand. Spacecraft rotational rates during deployment were captured at 2.2deg/s induced by the deployment of the boom – well within the capability of the spacecraft ADCS to compensate for.

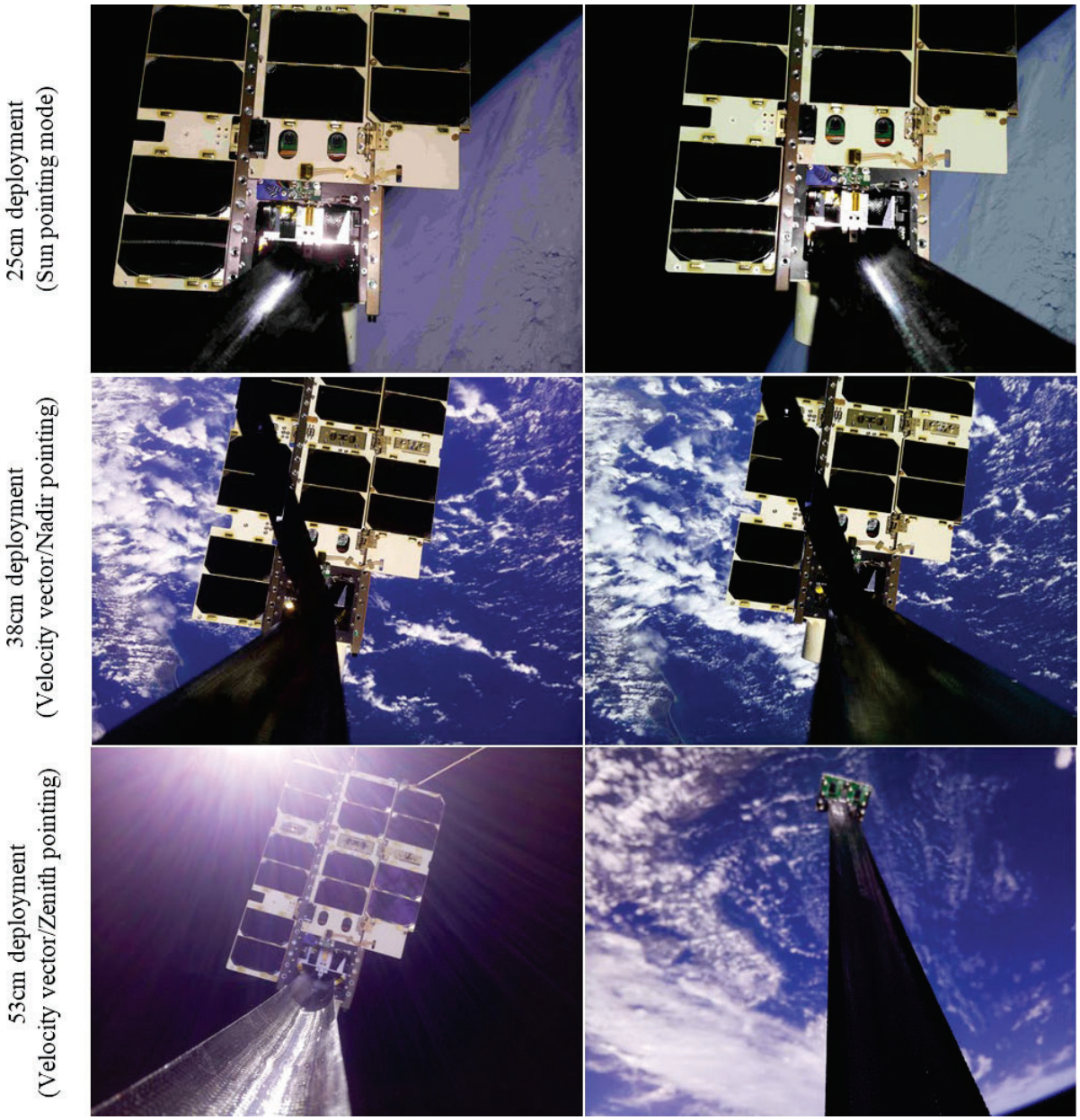


Figure 10: Stereo imagery from initial deployments of the Inspection boom (top, middle) and matched image from the boom tip camera and deployables camera (bottom)

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A further deployment to 53 cm was conducted with the spacecraft rolled to point the deployment direction (-X) towards Nadir. To achieve good lighting conditions, this was conducted passing over Northern Europe, close to a thermal minimum for the boom system at -15 °C as measured by the TMU.

After each deployment and stow operation, a further image was captured after stow to verify the telemetry feedback from home position microswitches.

Images captured from boom deployments are shown in Figure 10 and demonstrate the environmental impact on the outside of the spacecraft. These images verify full deployment of the solar arrays and antennas, as well as atomic oxygen and UV degradation of spacecraft materials. Finally, a small impact mark can be observed on one of the solar cells which was covered for launch by the deployable panel and not present in pre-launch photographs, strongly implying MMOD impact as a cause (Figure 11)

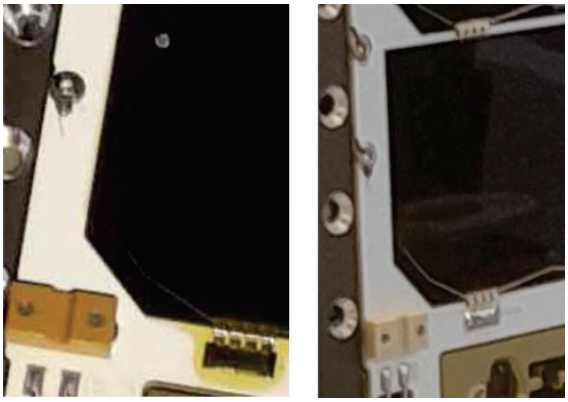


Figure 11: Zoomed in image from the inspection boom taken 2 months into mission (left) and pre-launch photograph of same area (right)

4.5 Education and outreach

A core element of the TPA-1 mission is its use in education and outreach. Since launch, the mission has garnered attention from the space community. The mission has been used as a training opportunity for a class of eight student interns, providing an opportunity to get hands on with operating a satellite and scheduling imaging operations. Visits during satellite passes by students and members of the public have seen over 150 having an opportunity to engage with the mission in its early stages. Postgraduate students have also placed requests for spacecraft tasking to support their research.

Imagery from the mission will be open sourced to the community via EO Lab Aotearoa, an organisation established to support use of Earth Observation data in New Zealand [8].

5 TPA-1 Future Operations and Applications

5.1 Imaging operations

At time of writing, the TPA-1 mission has been operating successfully for 2.5 months. Many further operations of the spacecraft are planned to support community outreach, education and engagement including school visits and “hackathons”.

Imaging operations are being planned to perform video capture, timelapses and to increase the level of detail available by increased quality in windowed mosaiced frames to still allow for timely download of high quality in specific areas of interest.

The imaging payload has been designed to support python code for image processing, and first demonstration of this capability is planned before the end of the year for wider role out as an educational tool in 2026.

5.2 Boom operations

Whilst the inspection boom is already returning interesting qualitative imagery of the condition of the spacecraft, more routine operations are planned to characterise colour changes and detect MMOD strikes to the solar cells. Timelapse and video captures will also be used to characterise the stability of the boom, both in its deployed state and during the process of deploying and stowing.

As with the Imagers payload, the boom tip payload hosts a Raspberry Pi compute module able to run scripting, this can be used as a test bed for stereoscopic pose and distance estimation algorithms. Moreover, the mission will generate a valuable dataset of imagery at a range of proximity distances, scene and lighting conditions with utility in development of rendezvous GNC capabilities.

5.3 Dragsail deployment

The final demonstration on the mission is the deployment of the dragsail at end of life. It is intended to perform a video capture of this deployment and to perform further imagery with the inspection boom. Communications and attitude control authority will be assessed throughout the descent once the sail has been deployed to determine capability to modulate the spacecraft drag through attitude changes.

5.4 Future applications

Given the already observed modulation of drag, experimental operation plans are being developed to perform a simulated collision avoidance manoeuvre, autonomously re-orienting the spacecraft to modulate its drag profile assessing predicted position from orbit propagators vs actual position established from GPS. Should communications and attitude control be well maintained after dragsail deployment, such experiments may be extended to low altitude, high drag manoeuvres.

Tools to be developed and implemented on the inspection boom are expected to find application for autonomous anomaly and change detection on board

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spacecraft, of particular utility to high cost long duration missions where servicing is considered, such as geostationary platforms or spacestations.

The TMU payload has proven to be a flexible baseline system for building other simple payloads and is enabling further student research projects to make rapid progress, including development of a radiation sensing payload.

5.5 Commercialisation

During the development of the mission, regulatory changes in the US reduced the permitted deorbiting time for satellites at end of mission from 25 years to 5 years. This regulatory forcing triggered formation of a spin-out company Frond Space Systems to commercialise the dragsail and boom system hosted on TPA-1. A combination of the two mechanisms resulted in a 10 m^2 MICRO sail system suitable for deorbiting microsatellites. The system has a mass of 1.6kg and utilises spring loaded BRC booms from RTL to deploy a metalised Kapton film sail[9]. A first system was procured by Gilmour Space Technologies and also launched on Transporter-14. The mission is underway with the MICRO dragsail awaiting deployment.

The mechanisms developed by TPA-SI and later by Frond have applications in deployable instrumentation (e.g.: magnetometers) and as support structures for other deployable systems such as solar arrays, an active area of research.

6 Discussion & Conclusions

The TPA-1 mission has successfully achieved its primary objectives in the initial months of operation, serving as a critical step in building New Zealand's sovereign space capability. The successful design, integration, launch, and operation of the CubeSat have validated the processes, facilities, and personnel at Te Pūnaha Ātea – Space Institute, fulfilling the mission's foundational goal as a capability development programme.

The restowable boom system has performed well, demonstrating reliable, multi-cycle deployment and providing rare inspection imagery for a satellite of this class. The identification of a potential MMOD impact on a solar cell highlights the practical utility of such a system for in-orbit diagnostics and health monitoring, a capability typically reserved for much larger platforms, though still rare in this context. The Earth observation payload has consistently returned high-quality imagery, proving the viability of ruggedised commercial-off-the-shelf hardware for accessible remote sensing applications. Its open, Python-based architecture has already been leveraged as an effective educational tool, engaging students in active mission operations.

A significant outcome of the TPA-1 programme extends beyond technical demonstration to successful commercialisation. The development of the compact dragsail and boom technologies directly led to the



Figure 12 Frond MICRO dragsail in stowed (top left) and deployed (bottom right) state and integrated onto the Gilmour Space ElaraSat Satellite[10] (right)

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formation of Frond Space Systems. This demonstrates a clear pathway from academic research and capability development to tangible economic and industrial impact within the national space ecosystem.

TPA-1 has demonstrated its core technologies and established an end-to-end mission framework to support future missions from NZ and international organisations. The data and operational experience gained will inform the development of future, more complex missions. The platform will continue to serve as a valuable in-orbit testbed for new algorithms, a tool for education and outreach, and a foundation for ongoing research and commercial ventures in the New Zealand space sector.

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